centroid can be inaccurate: this in turn lead to slower convergence, because some points can be assigned back and forth to different, almost equally-close centroids.

In summary, we need to keep in mind that k-means is a good option only for low-to-medium-dimensional (with at most around 20 dimensions) datasets where we know that clusters can be accurately approximated with hyperspheres.

12.2.3 K-means Performance Analysis

When we do know that a dataset meets the pre-conditions to apply k-means, however, this algorithm is a viable option and produces good results quickly. How quickly? That's what we are going to ascertain in this section.

Let's assume we have a dataset with `n` points, each point belonging to a `d`-dimensional space (hence each point can be represented as a tuple of `d` real numbers), and that we would like to partition the points into `k` different clusters.

If we examine each sub-step in listing 12.1, we can conclude that:

- The random initialization step takes `O(n*d)` assignments (`d` coordinates for each of the `n` points);
- Initializing cluster indices takes `O(n)` assignments;
- The main cycle is repeated `m` times, where `m` is the maximum number of iterations allowed;
- Assigning points to centroids takes `O(k*n*d)` operations, since for each point, we need to compute `k` `d`-